BUS 303 Signature Term Project - Deliverable Five

Starbucks vs. Dunkin' Donuts Competitive Analysis

Final Project Reflection and Summary

GitHub Repository: https://github.com/Moeedk40/BUS303-Starbucks-Project

D5.1 - Reflection

Throughout this project, I learned how Python and data analytics can be used to turn raw data into useful business information. At the beginning, the datasets were not ready for analysis. I had to inspect the data, identify missing values, duplicates, special characters, and incorrect data types, and then clean the information before I could make accurate comparisons. This helped me understand that data cleaning is an important part of the analysis process because incorrect or inconsistent data can lead to misleading conclusions.

One of the biggest challenges I faced was working through errors and understanding why certain parts of the code did not work as expected. Instead of only focusing on getting the code to run, I learned to examine the dataset, check column names and data types, and validate the results after making changes. Breaking larger problems into smaller steps helped me solve problems more effectively. I also learned the importance of checking my work after cleaning the data rather than assuming that a change worked correctly.

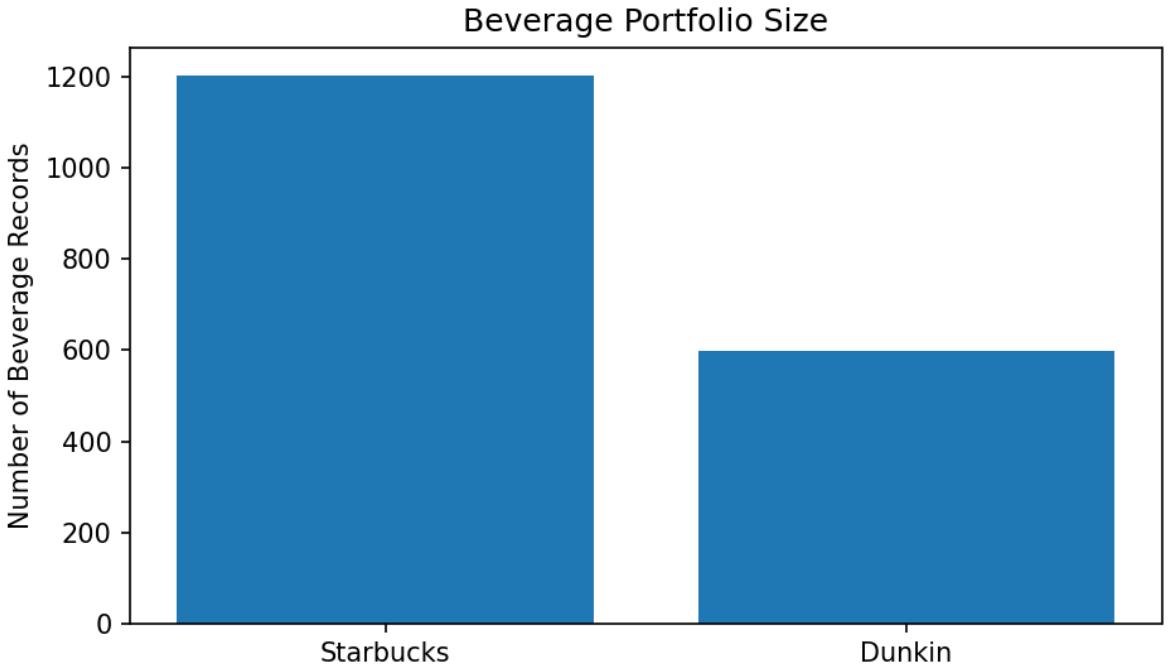
The competitive analysis between Starbucks and Dunkin' helped me understand how data can support business decisions. By comparing beverage portfolio size, average calories, sugar, protein, healthier choices, zero-calorie options, caffeine, and other nutritional information, I was able to identify differences between the two companies. The analysis showed that Starbucks had more beverage records and more options that met the healthier criteria used in the project, while Dunkin' had higher average calories and sugar in the dataset. Visualizations made these differences easier to understand and communicate.

Overall, this project improved my confidence using Python, pandas, Jupyter Notebook, and data visualization tools. I learned how to import, clean, organize, analyze, visualize, and interpret data instead of simply looking at numbers. More importantly, I learned how to connect the results of an analysis to real business questions. These skills can be applied to future business problems involving customer trends, products, financial information, or other datasets where decisions need to be supported by evidence.

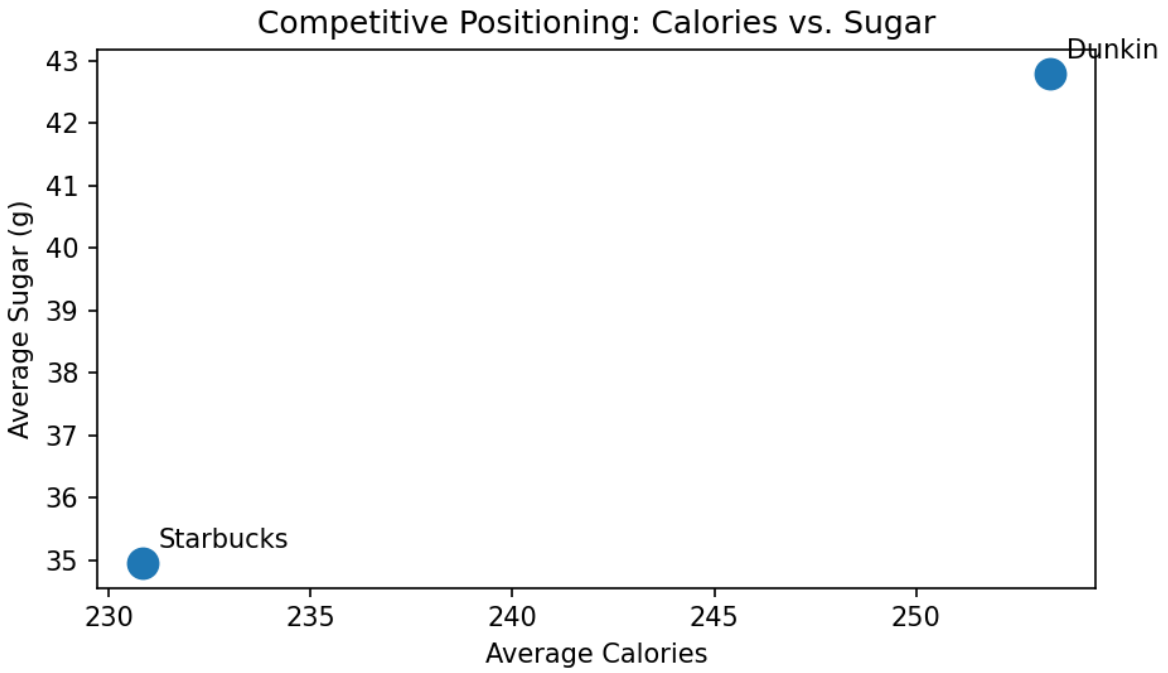
Final Competitive Analysis Summary

Metric	Starbucks	Dunkin
Beverage Records	1,203	599
Average Calories	230.83	253.33
Average Sugar (g)	34.95	42.79
Average Protein (g)	5.99	5.99

Metric	Starbucks	Dunkin
Healthier Choices	138	81
Zero-Calorie Records	40	0



Starbucks had 1,203 beverage records compared with 599 for Dunkin'. Based on the cleaned datasets, Starbucks offered a broader beverage portfolio and more product variation.



Dunkin' beverages averaged about 253.33 calories and 42.79 grams of sugar, compared with Starbucks at about 230.83 calories and 34.95 grams of sugar. In this dataset, Starbucks was positioned somewhat lower on both measures.

Healthier and Zero-Calorie Choices

Using the project definition of a healthier choice - 100 calories or less and 10 grams of sugar or less - Starbucks had 138 qualifying beverage records while Dunkin' had 81. Starbucks also had 40 zero-calorie records, while Dunkin' had none in the cleaned dataset. These findings suggest an advantage for Starbucks among customers specifically seeking lower-calorie and lower-sugar options.

Top 5 Most Caffeinated Starbucks Beverages

Beverage	Caffeine (mg)
Brewed Coffee - True North Blend Blonde Roast	475
Clover Brewed Coffee - Medium Roast	445
Clover Brewed Coffee - Light Roast	425
Brewed Coffee - Medium Roast	410
Clover Brewed Coffee - Dark Roast	380

The highest-caffeine Starbucks items were concentrated among brewed and Clover coffee products. The strongest option in the dataset was Brewed Coffee - True North Blend Blonde Roast at 475 mg.

Top 5 Most Sugary Dunkin' Beverages

Beverage	Sugar (g)
Butter Pecan Swirl Frozen Coffee with Whole Milk, Large	185
Butter Pecan Swirl Frozen Coffee with Skim Milk, Large	185
Butter Pecan Swirl Frozen Coffee with Cream, Large	181
Caramel Swirl Frozen Coffee with Skim Milk, Large	176
Caramel Swirl Frozen Coffee with Whole Milk, Large	175

The most sugary Dunkin' beverages were large frozen coffees, especially Butter Pecan Swirl and Caramel Swirl varieties. The highest recorded sugar content was 185 grams.

Business Conclusion

The analysis shows meaningful differences in how Starbucks and Dunkin' are positioned nutritionally. Starbucks had a larger beverage portfolio, more beverages meeting the project's healthier-choice criteria, and more zero-calorie options. Dunkin' had higher average calories and sugar and several very high-sugar frozen beverages. Protein was essentially identical on average across the two datasets.

From a competitive perspective, Starbucks may have an advantage with customers who value variety and lower-calorie or lower-sugar choices, while Dunkin' appears more concentrated in richer and sweeter beverage options. The project also demonstrates how cleaning, validating, visualizing, and interpreting data can turn menu information into useful competitive business insight.

Submission Materials

Jupyter Notebook (.ipynb), HTML copy, PDF copy, and GitHub repository link.